Connect:
Geometry and Algebra

Diego Rincón

Independent

Abstract

Geometry and algebra are two grammars reading the same persistence. Every major theorem in modern mathematics is a connection between them: the Langlands correspondence (Galois = automorphic), the $R = T$ theorem (deformation ring = Hecke algebra), the comparison isomorphisms ($H_{\mathrm{dR}} = H_{\mathrm{sing}}$, $H_{\text{ét}} = H_{\mathrm{crys}}$), the Weil conjectures (Frobenius eigenvalues = zeros), and BSD (rank of $E(\mathbb{Q})$ = order of vanishing of $L(E,s)$). In each case: a geometric object and an algebraic object are identified as the same thing. The identification is the content of mathematics [1]. The motive [2] is the universal connection: the object from which both geometry and algebra derive, the grammar behind all grammars. The visual cortex connects geometry and algebra in the body: it reads geometric structure (edges, orientations, shapes) and outputs algebraic invariants (representations of symmetry groups). Vision is the organism's version of the Langlands correspondence.

1 The Connection

Definition 1 (Connection between geometry and algebra)**.** A *connection* between geometry and algebra is an identification of a geometric object G and an algebraic object A as two descriptions of the same underlying structure M:

$$G \xleftarrow{\sim} M \xrightarrow{\sim} A.$$

The identification is not an approximation, not an analogy. It is an isomorphism: every property of G is a property of A, and vice versa. The connection reveals that geometry and algebra were always reading the same thing from different sides.

2 The Major Connections

Theorem 2 (The body of connections)**.** *The following are the major connections between geometry and algebra in the body of this work:*

Geometric side	*Algebraic side*	*The connection*
Variety $X/\mathbb{F}_q$	*Frobenius eigenvalues*	*Weil conjectures [3]*
Shimura variety $\mathrm{Sh}(G, X)$	*Automorphic forms*	*Eichler–Shimura [4]*
Galois representation ρ	*Modular form* f	*Modularity* ($R = T$) *[5]*
Selmer group H^1_f	*Mordell–Weil* $E(\mathbb{Q})$	*BSD [6]*
de Rham cohomology	*Singular cohomology*	*de Rham isomorphism [2]*
Étale cohomology	*Crystalline cohomology*	*Fontaine comparison [7]*
Fargues–Fontaine curve	G_K*-representations*	*Fargues–Scholze [7]*
Hitchin fibration	*Orbital integrals*	*Fundamental lemma [8]*
Motive $h^*(X)$	*All cohomologies*	*Universal connection [2]*

In every row: a geometric object on the left, an algebraic object on the right, and the theorem that identifies them. The motive is the last row: not a specific connection but the universal one, the object behind all the others.

3 The Visual Cortex as Connection

Proposition 3 (The brain's Langlands correspondence). *The visual cortex connects geometry and algebra in the body.*

Geometric input: the visual field is a two-dimensional manifold, a map $f : \mathbb{R}^2 \to \mathbb{R}$ *(luminance at each point). The geometry is local: edges, gradients, curvature, texture.*

Algebraic output: the visual cortex extracts invariants — representations of the symmetry group of the image. A Gabor filter at orientation θ *detects the component of the image transforming under rotation by* θ*: it computes the projection onto the* θ*-isotypic component of the action of* $SO(2)$ *on the image. This is representation theory.*

The visual cortex performs the Langlands correspondence locally: it reads geometric structure (the image) and extracts algebraic invariants (the symmetry representations). V1 sees edges. V2 sees combinations. V4 sees shapes. IT sees objects. Each level is a higher algebraic invariant of the same geometric input. The hierarchy is the connective tower [5]: each floor adds one more level of algebraic structure to the geometric data.

Vision is the organism's version of cohomology: reading what persists in the image under the transformations the organism undergoes (movement, lighting change, viewpoint shift). The persistent invariants are the algebraic output. The geometry is the input. The visual cortex is the connection.

4 Connective Sequences

Proposition 4 (The Postnikov tower as connective sequence). *A connective* E_∞*-ring* R *[5] has a Postnikov tower:*

$$R \to \cdots \to \tau_{\leq 2} R \to \tau_{\leq 1} R \to \tau_{\leq 0} R,$$

where $\tau_{\leq n} R$ *is the* n*-truncation:* $\pi_i(\tau_{\leq n} R) = \pi_i(R)$ *for* $i \leq n$ *and* 0 *for* $i > n$*.*

Each truncation $\tau_{\leq n} R$ *is a connection at level* n*: it captures the geometric-algebraic relationship up to* n*-th order obstruction.* $\tau_{\leq 0} R = \pi_0(R)$ *is the classical connection — the* *classical ring $R(\bar{\rho})$, the Hecke algebra $\mathbb{T}$. $\tau_{\leq 1}R$ adds the first derived correction — the derived Hecke operators of Venkatesh [5]. The full R is the complete connection — geometry and algebra identified at every level simultaneously.*

The connective sequence is the connection made precise: not just an isomorphism at one level but a tower of isomorphisms, each floor a deeper identification, the limit the full derived equivalence.

5 Algebra as Fixed Geometry

Proposition 5 (Algebraic statues)**.** *An algebraic object is a geometric object that has been fixed by a group action: the Galois-invariant classes in $H^*_{\text{ét}}(X_{\bar{k}})$ are the algebraic cycles (Tate conjecture [9]). The fixed points of Frobenius on $H^*_{\text{ét}}$ are the eigenvalues — the algebraic data of the Weil conjectures.*

Algebra is geometry that does not move under the symmetry. A statue is a sculpture that has been fixed: carved from the geometric material, invariant under the transformations that change everything else. The algebraic objects are the statues of geometry — what remains when the geometric symmetry acts and the object does not change.

The Langlands correspondence is the statement that the statues of geometry (Galois representations) and the statues of algebra (automorphic representations) are the same statues, seen from opposite sides.

6 The Universal Connection

Theorem 6 (The motive as universal connection)**.** *The motive $h^*(X)$ [2] is the universal connection between geometry and algebra. It is the object M in*

$$\textit{geometry of } X \xleftarrow{\sim} h^*(X) \xrightarrow{\sim} \textit{algebra of } X,$$

from which every specific connection derives by change of coefficients, by realization functor, by specialization to a particular grammar.

Every row in Theorem 2 is a realization of the motive: a specific identification obtained by reading $h^*(X)$ through one geometric grammar and one algebraic grammar simultaneously. The comparison isomorphisms are the statement that different geometric and algebraic grammars give the same reading of the motive.

The motive is the object behind the connection. The connection is what the motive looks like when two grammars discover each other. Everything is everything [1]: every connection is the motive, read twice.

Remark 7 (Vision and the motive)*.* The visual cortex reads the motive of the scene: the underlying object — shape, material, identity — that persists through all viewpoints, all lighting conditions, all distances. The geometric input (pixels, edges, gradients) and the algebraic output (object identity, category, relation) are two realizations of the scene's motive.

Seeing is finding the motive. Mathematics is proving it exists. Both are the same practice at different scales of precision.

References

[1] Rincón, D. (2026). Mathematics. Preprint.

[2] Rincón, D. (2026). Motives, Standard Conjectures, and the Unification of L-functions. Preprint.

[3] Rincón, D. (2026). The Weil Conjectures as Proof of Concept for RH. Preprint.

[4] Rincón, D. (2026). Shimura Varieties: The Geometry Behind the L-Functions. Preprint.

[5] Rincón, D. (2026). Galois Deformation Theory and Derived Algebra. Preprint.

[6] Rincón, D. (2026). BSD via Arithmetic Gan–Gross–Prasad. Preprint.

[7] Rincón, D. (2026). Perfectoid Spaces and the Local Frobenius. Preprint.

[8] Rincón, D. (2026). Automorphic Forms and the Trace Formula. Preprint.

[9] Rincón, D. (2026). The Tate Conjecture: Arithmetic Hodge and the Galois Obstacle. Preprint.